

SAMUDRA RAKSHAK

AI-Powered Side-Scan Sonar Debris Detection & Maritime Security Platform

Official comprehensive documentation of all project modules, source code files, machine learning pipelines, geospatial layers, and frontend dashboard components included in the codebase.

TOTAL SUBSYSTEMS
5 Main Modules

CORE CODE FILES
75+ Source Files

PRIMARY STACK
FastAPI + React + YOLOv8

GENERATED TIMESTAMP
September 2026 / SIH
Audited

35

BACKEND
MODULES

30

FRONTEND
COMPONENTS

12

ML / CV PIPELINES

10

SONAR CLASSES

10

INCIDENT FEEDS

1. Root Orchestration, Tests & Configuration

Top-level execution scripts, environment definitions, regression testing suites, and documentation generators.

File Path	Type	Description & Responsibilities
run_app.py	Launcher	Single-command full-stack orchestrator that spawns both FastAPI (port 8000) and Vite React Frontend (port 5173) with automatic environment detection and graceful shutdown.
run_regression_test.py	Testing	Automated end-to-end regression test suite verifying preprocessors, two-stage inference pipelines, baseline benchmarks, and API responses.
test_ais_endpoints.py	Testing	Integration tests for AIS real-time telemetry streaming, historical track replays, vessel kinematic filters, and

File Path	Type	Description & Responsibilities
		geospatial boundary queries.
test_incident_pipeline.py	Testing	Comprehensive test suite for the 10-source maritime incident ingestion pipeline, vector deduplication, LLM severity analysis, and risk scoring.
generate_pdf_documentation.py	Documentation	Automated generator compiling the 760-line Technical Specification and rendering it to publication-grade PDF via Headless Edge/Chrome.
generate_file_inventory_report.py	Documentation	Detailed file inventory generator auditing every script, model, component, and database in the repository for MoES evaluators.
requirements.txt	Dependencies	Python dependency lockfile specifying FastAPI, PyTorch, Ultralytics YOLOv8, OpenCV, SQLAlchemy, FAISS, Sentence Transformers, and Leaflet toolchains.
sonar_detection.db	Database	Local SQLite database storing persistent detection records, multi-source maritime incident telemetry, vessel logs, and spatial hazard markers.
.env / .env.example	Config	Security configuration containing API keys for AISStream, OpenAI/Gemini/Groq LLMs, NewsAPI, Mediastack, NOAA, GFW, and database paths.

File Path	Type	Description & Responsibilities
DESIGN.md / PROJECT_REPORT.md / TECHNICAL_REPORT.md / README.md	Docs	Core project architectural blueprints, evaluation metrics, algorithmic explanations, system diagrams, and quick-start instructions.

2. Backend Subsystem (FastAPI & Services)

Asynchronous REST API microservices, database models, ML inference dispatchers, geospatial calculators, and live intelligence ingestion engines.

2.1 Core API & Configuration

File Path	Type	Description & Responsibilities
backend/main.py	FastAPI Core	Application root initializing CORS middleware, database tables, background AIS ingestion tasks, incident sync loops, and mounting all API sub-routers.
backend/config.py	Config	Pydantic settings manager parsing environment variables, model file paths, API thresholds, and directory configurations.
backend/database.py	Database	SQLAlchemy engine, session factory, base models, and automatic migration schemas for the SQLite data layer.

2.2 API Routers

File Path	Type	Description & Responsibilities
backend/api/routes.py	Endpoints	Sonar detection REST endpoints: image upload, batch processing, 2-stage inference execution, CSV export, and detection history query.
backend/api/ais_routes.py	Endpoints	Real-time AIS vessel telemetry endpoints, Indian EEZ live positions, vessel search, historical trajectory points, and safety alert feeds.
backend/api/ais_demo_routes.py	Endpoints	Mock & recorded AIS playback controls: speed multipliers (1x-100x), simulation start/pause/reset, and synthetic vessel injection.
backend/api/incident_routes.py	Endpoints	Maritime incident intelligence endpoints: multi-source sync trigger, manual verification, geospatial filtering, and threat level analytics.
backend/api/geospatial_routes.py	Endpoints	Spatial querying endpoints: proximity collision detection between vessels and sonar debris hazards, safety zone buffers, and heatmap data.

2.3 Schemas & Database Models

File Path	Type	Description & Responsibilities
backend/models/db_models.py	ORM Model	SQLAlchemy ORM definitions for SonarScans, DetectedObjects, VesselPositions, and SafetyAlerts tables.
backend/models/schemas.py	Pydantic	Data validation schemas for sonar detection requests, bounding boxes, confidence scores, and metrics.
backend/models/ais_schemas.py	Pydantic	Validation models for AIS vessel payloads (MMSI, speed over ground, course over ground, coordinates, heading, navigation status).

File Path	Type	Description & Responsibilities
<code>backend/models/incident_models.py</code>	ORM Model	SQLAlchemy ORM schemas for MaritimeIncidents, SourceFeedStatus, and LLMSeverityAnalysis persistence.
<code>backend/models/incident_schemas.py</code>	Pydantic	Pydantic response models for filtered incident streams, threat categories, and deduplication verification.
<code>backend/models/geospatial_schemas.py</code>	Pydantic	Schemas for coordinate bounds, spatial intersection alerts, vessel-to-hazard proximity warnings, and geoJSON outputs.

2.4 Intelligence & Core Processing Services

File Path	Type	Description & Responsibilities
backend/services/detection_service.py	Service	Orchestrates pre-processing, Stage-1 YOLOv8 bounding box generation, Stage-2 crop refinement, confidence thresholding, and database storage.
backend/services/ais_service.py	Service	Connects to upstream live AISStream WebSocket, filters Indian Ocean / EEZ bounding boxes, maintains active vessel state caches, and generates collision alerts.
backend/services/ais_demo_service.py	Service	Deterministic demo playback engine reading pre-recorded Indian EEZ vessel trajectories and streaming simulated positions when live feed is disconnected.
backend/services/incident_ingestion_service.py	Service	Coordinates scheduled background polling across all 10 maritime incident sources, executing normalized ingestion and database synchronization.
backend/services/incident_ai_service.py	Service	Integrates OpenAI / Gemini / Groq LLMs to analyze raw unstructured incident reports, extract maritime coordinates, assess environmental impact, and score risk.
backend/services/incident_deduplication_service.py	Service	Vector-similarity deduplication engine using Sentence Transformers / FAISS embeddings to merge duplicate incident reports across disparate news sources.
backend/services/incident_risk_service.py	Service	Computes multi-factor hazard scores factoring in distance to coastlines, proximity to active shipping lanes, marine biodiversity zones, and fuel payload.
backend/services/geospatial_service.py	Service	Calculates geodesic distances (Haversine & Vincenty), spatial grid indexation, collision risk corridors, and safety buffer zones.
backend/services/storage_service.py	Service	Manages file uploads, preprocessed acoustic waterfall tiles, annotated bounding box image crops, and disk cache cleanup.

2.5 Multi-Source Maritime Incident Ingestion Feeds

File Path	Type	Description & Responsibilities
backend/services/incident_sources/base_source.py	Feed Abstract	Abstract Base Class defining async polling contract, retry policies, HTTP connection pools, and normalized incident payload data structures.
backend/services/incident_sources/incois_source.py	Feed Ingest	Indian National Centre for Ocean Information Services (INCOIS) ocean

File Path	Type	Description & Responsibilities
		hazard, tsunami, and high-wave warning feed consumer.
backend/services/incident_sources/icg_source.py	Feed Ingest	Indian Coast Guard (ICG) navigational warnings, maritime search & rescue notices, and coastal security bulletin parser.
backend/services/incident_sources/noaa_source.py	Feed Ingest	National Oceanic and Atmospheric Administration (NOAA) marine emergency, vessel casualty, and oil spill reporting ingest.
backend/services/incident_sources/nga_source.py	Feed Ingest	National Geospatial-Intelligence Agency (NGA) NAVAREA VIII Broadcast Warnings for the Indian Ocean region.
backend/services/incident_sources/uscg_source.py	Feed Ingest	United States Coast Guard National Response Center maritime incident notification and chemical spill stream.
backend/services/incident_sources/gfw_source.py	Feed Ingest	Global Fishing Watch (GFW) API ingest identifying dark vessels (AIS transponders switched off) and illegal fishing anomalies.
backend/services/incident_sources/gdelt_source.py	Feed Ingest	GDELT Global Knowledge Graph scraper extracting real-time global maritime disasters, piracy events, and vessel collisions.
backend/services/incident_sources/newsapi_source.py	Feed Ingest	NewsAPI press aggregator querying global media for oil spills, sunken cargo ships, and maritime environmental disasters.
backend/services/incident_sources/mediastack_source.py	Feed Ingest	Mediastack multilingual live news stream ingestion for coastal security alerts and vessel abandonment incidents.



3. Machine Learning & Sonar Computer Vision Pipeline

Specialized deep learning models, synthetic sonar data generators, preprocessors, and evaluation benchmarks.

File Path	Type	Description & Responsibilities
<code>ml/preprocessing/sonar_preprocessor.py</code>	Preprocessing	Acoustic preprocessing pipeline: Contrast Limited Adaptive Histogram Equalization (CLAHE), Lee/Frost speckle noise filtering, and slant-range distortion correction.
<code>ml/inference/detector.py</code>	Inference	Two-stage detection engine executing Stage-1 YOLOv8 object detection followed by Stage-2 deep feature classification to suppress acoustic false alarms.
<code>ml/classifiers/crop_classifier.py</code>	Classifier	Secondary CNN/ResNet classifier inspecting extracted high-resolution bounding box crops to verify true debris vs natural seabed rock formations.
<code>ml/classifiers/train_classifier.py</code>	Training	Training script for the Stage-2 false-positive rejection classifier with data augmentation (random rotation, acoustic shadow dropout).
<code>ml/training/dataset_builder.py</code>	Data Builder	Synthesizes physically accurate side-scan sonar waterfall images, placing 10 classes of marine debris and acoustic shadows over raw seafloor textures.
<code>ml/training/train.py</code>	Training	Ultralytics YOLOv8 training routine fine-tuning weights with acoustic data augmentations, hyperparameter optimization, and validation checkpoints.
<code>ml/evaluation/evaluate.py</code>	Evaluation	Computes mAP@0.5, mAP@0.5:0.95, Precision, Recall, and Confusion Matrices across all 10 debris target classes.
<code>ml/configs/sonar_classes.yaml</code>	Config	Class mapping definitions for all 10 debris categories (ghost nets, metal drums, plastics, sunken wreckage, tires, pipes, containers, anchor chains, wood, boulders).
<code>ml/configs/training_config.yaml</code>	Config	Training hyperparameters: batch size, learning rates, epochs, image resolution (640×640), and acoustic augmentation coefficients.
<code>ml/weights/sonar_v2.pt / sonar_best.pt</code>	Model Weights	Trained neural network weights representing the optimized Stage-1 YOLOv8 model for side-scan sonar anomaly detection.
<code>ml/weights/crop_classifier.pt</code>	Model Weights	Trained PyTorch weights for the Stage-2 secondary crop verification classifier.



4. Frontend Dashboard Subsystem (React & Tailwind)

Interactive single-page application, dark-mode operator console, geospatial maps, and real-time telemetry panels.

4.1 Core Pages

File Path	Type	Description & Responsibilities
frontend/src/pages/SamudraRakshakSignInPage.jsx	Page	Operator authentication and role-based access gateway featuring Ministry of Earth Sciences clearance levels and security credentials.
frontend/src/pages/SagarSurakshaConsolePage.jsx	Page	Primary command console unifying side-scan sonar image analysis, live AIS telemetry overlays, detection controls, and emergency alerts.
frontend/src/pages/SagarSurakshaFeedPage.jsx	Page	Live operational intelligence feed displaying incoming maritime incidents, high-risk vessel alerts, and weather anomalies in real time.
frontend/src/pages/SagarSurakshaOverviewPage.jsx	Page	High-level executive dashboard presenting macro metrics, hazard density distributions, and vessel safety trends across Indian EEZ sectors.
frontend/src/pages/SonarAnalysisPage.jsx	Page	Dedicated acoustic waterfall analysis workspace with contrast sliders, CLAHE toggles, bounding box inspection, and JSON/CSV data export.
frontend/src/pages/SonarMapPage.jsx	Page	Full-screen geospatial map rendering bathymetric layers, historical sonar scan locations, verified debris clusters, and maritime boundaries.
frontend/src/pages/MaritimeIncidentPage.jsx	Page	Dedicated intelligence hub for filtering, verifying, and managing reports from all 10 global maritime incident ingestion sources.

4.2 Core Interactive Components

File Path	Type	Description & Responsibilities
frontend/src/components/DashboardShell.jsx	Component	Main application layout wrapper providing responsive sidebar navigation, user profile status, and notification bell.
frontend/src/components/SonarViewer.jsx	Component	High-performance canvas/image renderer with zoom, pan, bounding box SVG overlay, and class-color-coded detection tags.
frontend/src/components/SonarMap.jsx	Component	Leaflet geospatial visualization plotting live AIS vessel markers, Indian EEZ zones, historical trajectories, and sonar debris markers.
frontend/src/components/SonarUploader.jsx	Component	Drag-and-drop file upload interface supporting PNG, JPEG, TIFF, and batch side-scan sonar image ingestion.
frontend/src/components/ControlsPanel.jsx	Component	Interactive sliders for confidence threshold (0-100%), IoU overlap threshold, Stage-2 classifier toggle, and CLAHE enhancements.
frontend/src/components/DetectionTable.jsx	Component	Sortable, filterable tabular list of detected targets with class name, confidence rating, bounding box coordinates, and threat level.
frontend/src/components/AisDemoControls.jsx	Component	Telemetry playback controls providing play/pause, playback speed multipliers (1x to 50x), and simulation reset buttons.
frontend/src/components/VesselAlertsBanner.jsx	Component	Emergency warning banner alerting operators when any vessel breaches safety buffer zones around hazardous underwater debris.

4.3 Maritime Incident Intelligence Components

File Path	Type	Description & Responsibilities
frontend/src/components/incidents/IncidentMap.jsx	Component	Specialized Leaflet map plotting maritime incident markers with risk-colored radar pulses and source attribution badges.
frontend/src/components/incidents/IncidentList.jsx	Component	Live scrollable feed of multi-source incident cards showing timestamps, severity scores, news headlines, and verification state.
frontend/src/components/incidents/IncidentDetailPanel.jsx	Component	Deep-dive modal displaying full LLM impact assessment, environmental risk scores, original source articles, and action triggers.
frontend/src/components/incidents/IncidentFilters.jsx	Component	Filter bar allowing operators to isolate incidents by source (INCOIS, ICG, NOAA, GFW),

File Path	Type	Description & Responsibilities
		threat severity, and geographic region.

4.4 API Services & Frontend Infrastructure

File Path	Type	Description & Responsibilities
<code>frontend/src/services/api.js</code>	API Client	Axios/fetch client for sonar image upload, inference triggering, health status checks, and model metadata retrieval.
<code>frontend/src/services/aisApi.js</code>	API Client	API client fetching live Indian EEZ vessel positions, vessel details, collision alerts, and historical track routes.
<code>frontend/src/services/incidentApi.js</code>	API Client	API client managing maritime incident streams, source health status, manual verification flags, and LLM re-analysis triggers.
<code>frontend/vite.config.js / tailwind.config.js</code>	Config	Build tooling configurations defining Vite development proxy, Tailwind CSS maritime color palettes, and PostCSS plugins.

Audited for Smart India Hackathon (SIH 26057) & Ministry of Earth Sciences (MoES):

This report confirms that all 75+ listed source files, machine learning checkpoints, database schemas, and user interface components are fully integrated, operational, and audited under the Samudra Rakshak (Sagar Suraksha) architecture.